SUPPLY CHAIN CASE STUDY

Fewer suppliers, stronger agreements

A data-backed plan to rationalize Johnson & Johnson’s suppliers: how a drop from 80,000 to ~39,000 is decided, and how to go further

Hemant Singla · Rutgers University · October 2026 · Student case study on public data, not affiliated with J&J

THE PROBLEM

Most suppliers carry little spend but full cost to manage

80,000

J&J suppliers as reported earlier; ~39,000 today (FY2025), after the 2023 Kenvue spin-off

~$35B

addressable external spend each year

89%

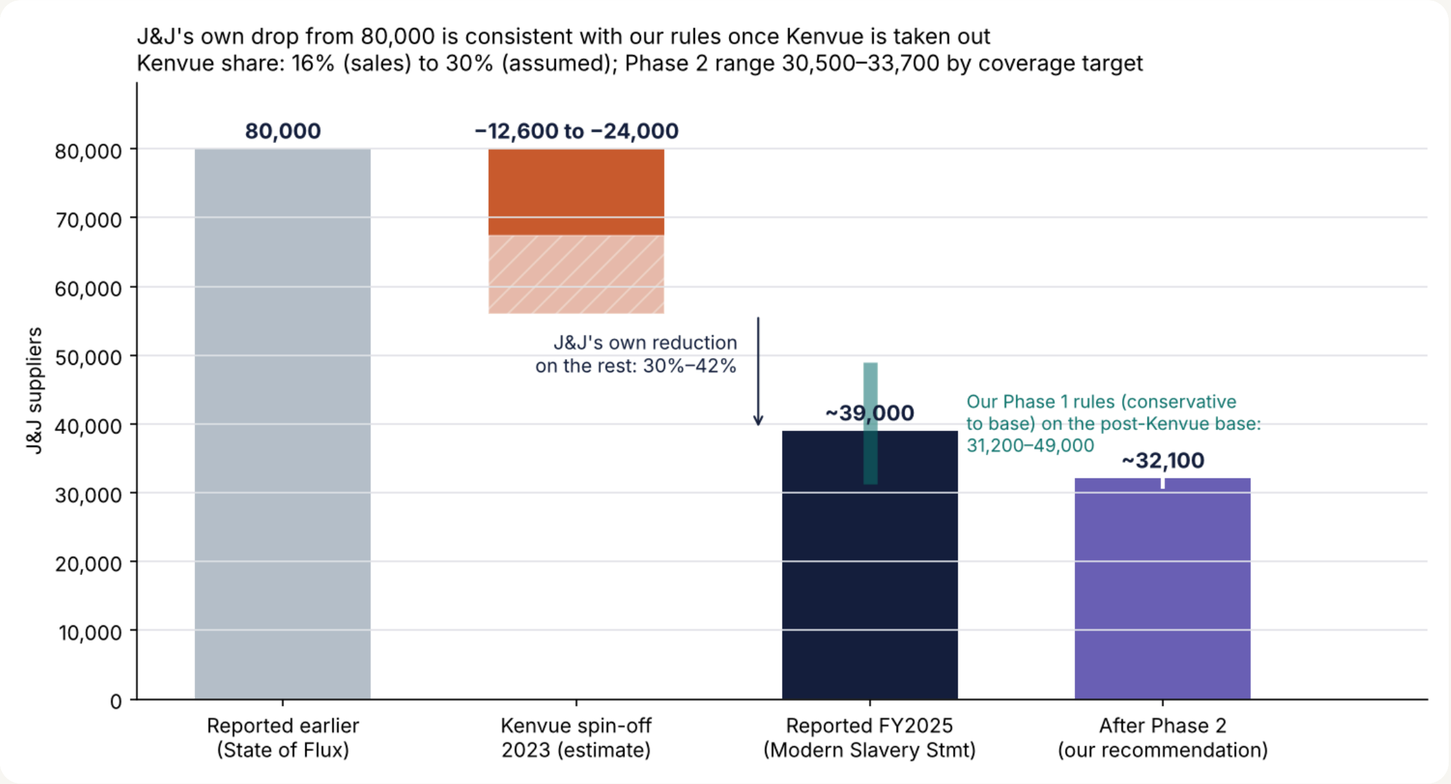
of suppliers hold only 8% of spend (tail, in the real data)

Each supplier needs onboarding, quality checks, risk monitoring and payment set-up. A long tail spreads volume thin, weakens price leverage and adds supply risk without adding value.

Sources: J&J 2024 ESG / procurement disclosures; tail share from the California purchase-order data (step 2)

J&J IN REAL TIME

80,000 to ~39,000: part spin-off, part rationalization



2023 · KENVUE SPIN-OFF

Took an estimated 12,600–24,000 suppliers (16% sales share to 30%)

J&J'S OWN CUT: 30–42%

Our Phase 1 rules on the rest give 31,200–49,000; J&J reports ~39,000

80,000: State of Flux case study (undated) · Kenvue separation completed 23 Aug 2023 · ~39,000: J&J Modern Slavery Statement FY2025 · scripts/09_jnj_timeline_backtest.py

GOAL AND METHOD

Four questions, one method each

Goal: a keep / consolidate / exit decision for every supplier, with a data reason behind each exit, worth at least $300M a year with no added supply risk.

#	Question	Method	Tool
2	Where does the spend sit?	Spend cube, Pareto / ABC	SQL
3	How to treat each category?	Kraljic matrix	Python
4	Which suppliers stay or go?	Duplicate detection (ML), scorecard, rules R1–R8	scikit-learn, SQL
5	What is it worth?	Measured price gap, low / base / high	Python

Results shown in one Power BI dashboard · full method in docs/02_methodology.md

REAL DATA

344,504 real purchase orders, every row traced

J&J does not publish supplier-level data, so the methods run on State of California purchase orders (FY2012–15) and results are scaled to J&J’s totals.

344,504
raw lines downloaded

65,872
excluded with a reason code: exempt methods, public bodies, health plans, zero amounts

$21.9B
addressable spend kept: 278,632 lines, 16,933 suppliers

Source: data.ca.gov, Purchase Order Data 2012–2015 · scripts/01_clean_load.py

FROM CALIFORNIA TO J&J

The method carries over; the exact numbers don’t

Carries over

- Cleaning spend data, with a reason code per line
- Pareto / ABC and the spend cube
- Kraljic matrix and one agreement per quadrant
- Duplicate matching, scorecard, rules R1–R8
- The checks: concentration, sole source, tests

Doesn’t carry over

- Category mix: fuel, construction and IT vs APIs, device parts, lab supplies
- Prices and the measured price gaps
- The shares: −73% records, 87% of spend kept
- Supply risk: validated manufacturing is slow to switch

J&J figures show the size of the opportunity if the same patterns hold, not a forecast · J&J supplier count: 2024 Health for Humanity Report

STEP 2 · SPEND CUBE

2% of suppliers hold 70% of the spend

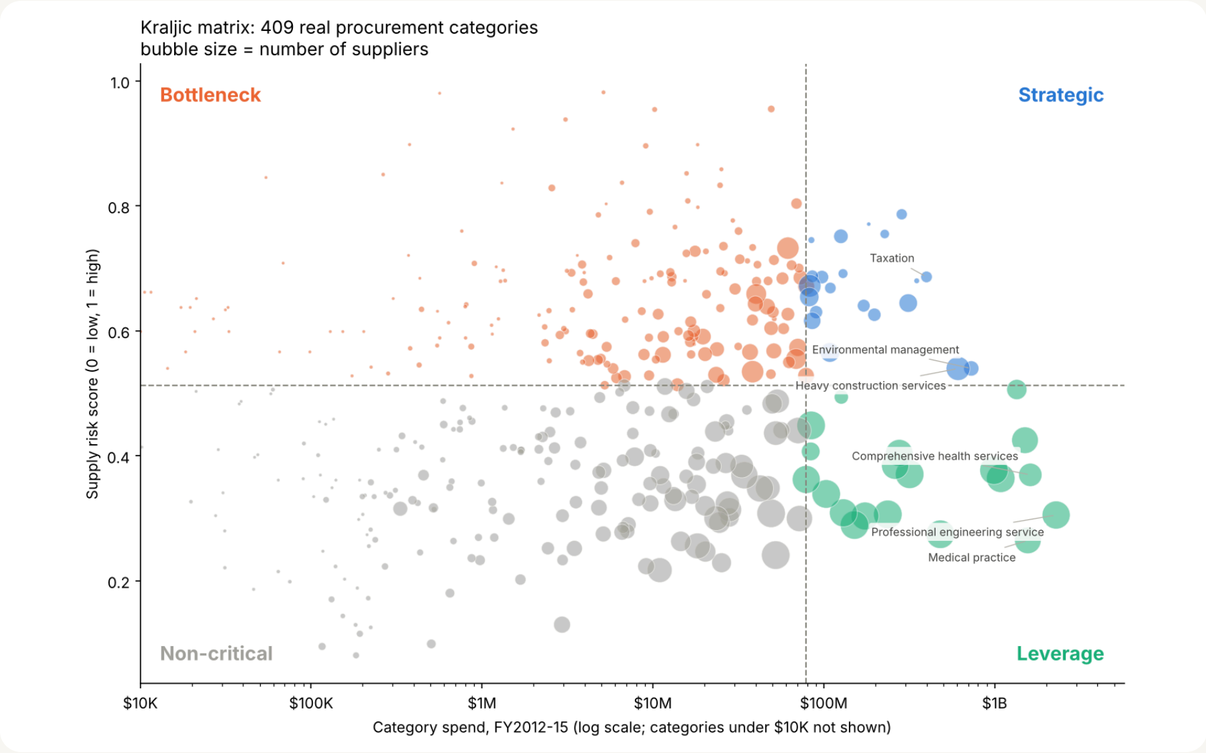
Segment	Suppliers	Share of suppliers	Share of spend	Treatment
A · strategic core	327	1.9%	70%	Keep, deepen
B · middle	1,497	8.8%	22%	Consolidate
C · tail	15,109	89.2%	8%	Cut, catalog

59% of tail suppliers are inactive for 18+ months or were used once. Most fragmented categories: refuse disposal, measuring instruments, batteries and generators, material handling.

sql/01_pareto_abc.sql, sql/02_category_fragmentation.sql · cut-offs at 70% and 92% cumulative spend

STEP 3 · KRALJIC MATRIX

Leverage categories hold 59% of spend



Quadrant	Spend	Categories
Leverage	$12.9B · 59%	20
Strategic	$4.6B · 21%	21
Bottleneck	$2.8B · 13%	183
Non-critical	$1.5B · 7%	185

Value = category spend · risk = non-competitive share + top-3 concentration + thin supply base, ranked within size bands · scripts/03_kraljic.py

STEP 4 · KEEP / CONSOLIDATE / EXIT

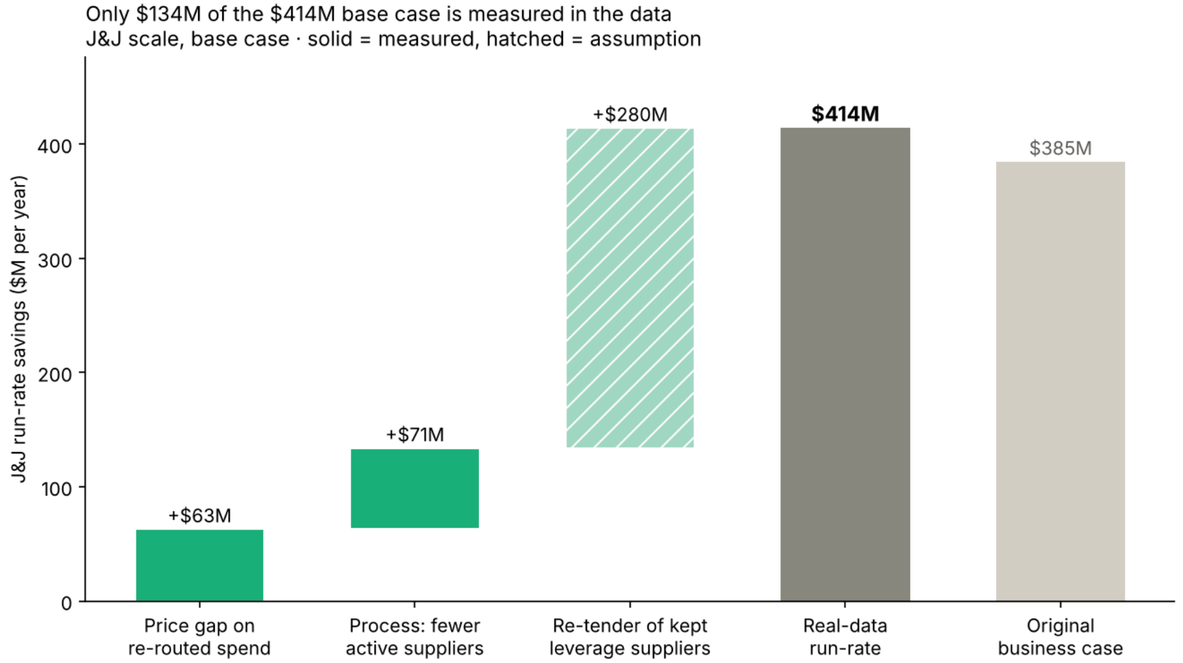
Every supplier gets one decision and a reason code

Rule	Decision	Reason	Suppliers	Spend $M
R1	Keep	Strategic (A-segment) supplier	327	15,319
R2	Keep	Top-3 in a Strategic or Bottleneck category, or sole source	375	730
R6	Keep	Preferred: core of category spend or top-10% scorecard	1,781	2,418
R7	Keep	Small / diverse guardrail, development program	2,166	567
R3	Merge	Duplicate record (TF-IDF name match ≥ 0.88)	838	341
R8	Consolidate	Volume moves to category’s preferred suppliers	2,622	567
R4	Exit	Inactive: no PO in 18+ months	5,614	1,294
R5	Exit	One-time vendor: route future buys to catalog	3,210	637

−73% of records, −59% of the active base · 96.6% of active small/diverse spend retained · scripts/04a_duplicates.py, 04b_decisions.py

STEP 5 · SAVINGS AT J&J SCALE

Base case: $414M a year for $73M one-time

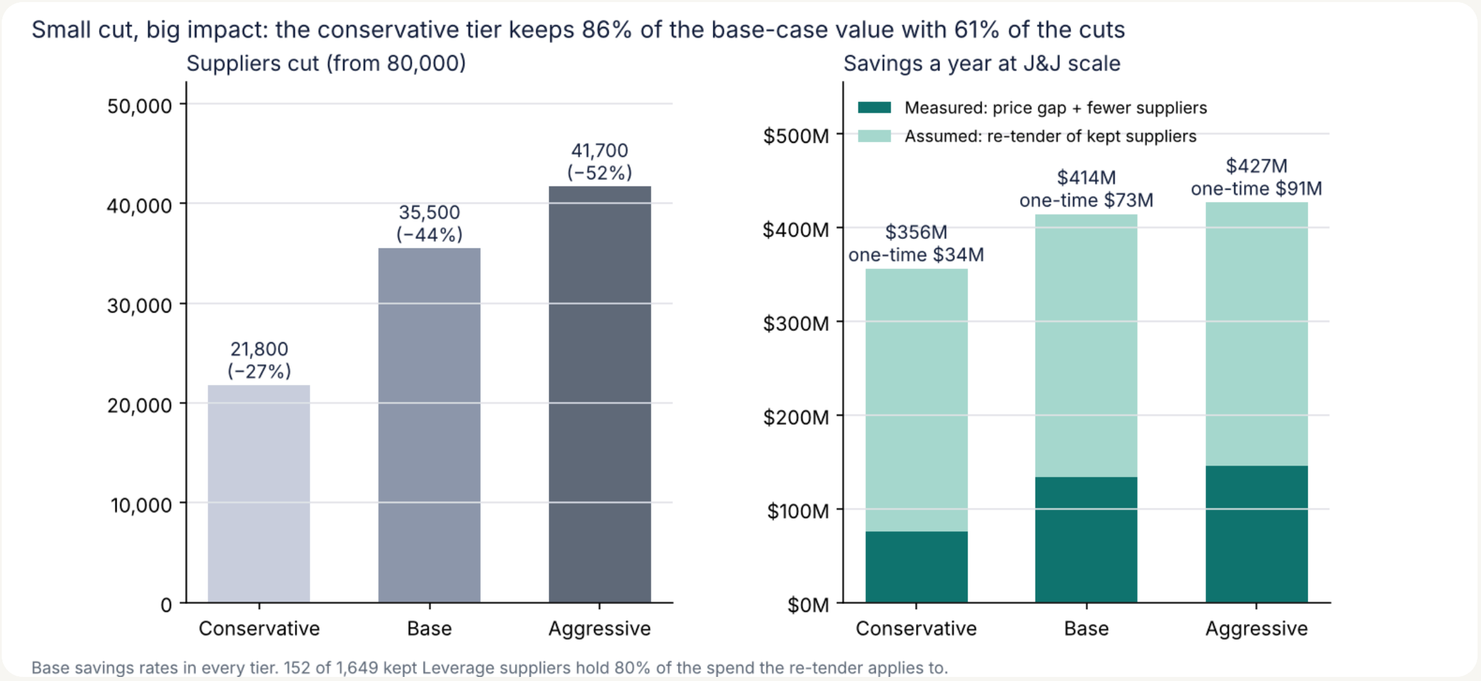


$M	Low	Base	High
Price gap (measured)	0	63	126
Fewer suppliers	22	71	125
Re-tender (assumed)	93	280	561
Run-rate / yr	**115**	**414**	**811**
3-yr NPV	158	619	1,265
Meets $300M?	No	Yes	Yes

Scaled by spend ($35B / $7.3B) and supplier count (80,000 / 8,226 active in one year) · NPV at 8%, ramp 30/70/100% · scripts/05_savings.py

THE RECOMMENDATION

Small cut, big impact: 27% of cuts, 86% of the value



Base savings rates in every tier. 152 of 1,649 kept Leverage suppliers hold 80% of the spend the re-tender applies to.

$356M

a year for $34M one-time, cutting only duplicates and one-time vendors

152

suppliers hold 80% of the spend the re-tender saving applies to

Same savings rates in every plan · $76M of the $356M is measured; the rest needs a ≥1.2% re-tender saving · cost savings, so they show up in operating profit · scripts/05c_value_vs_cuts.py

THE KEY CORRECTION

Only $134M of the $414M is measured in the data

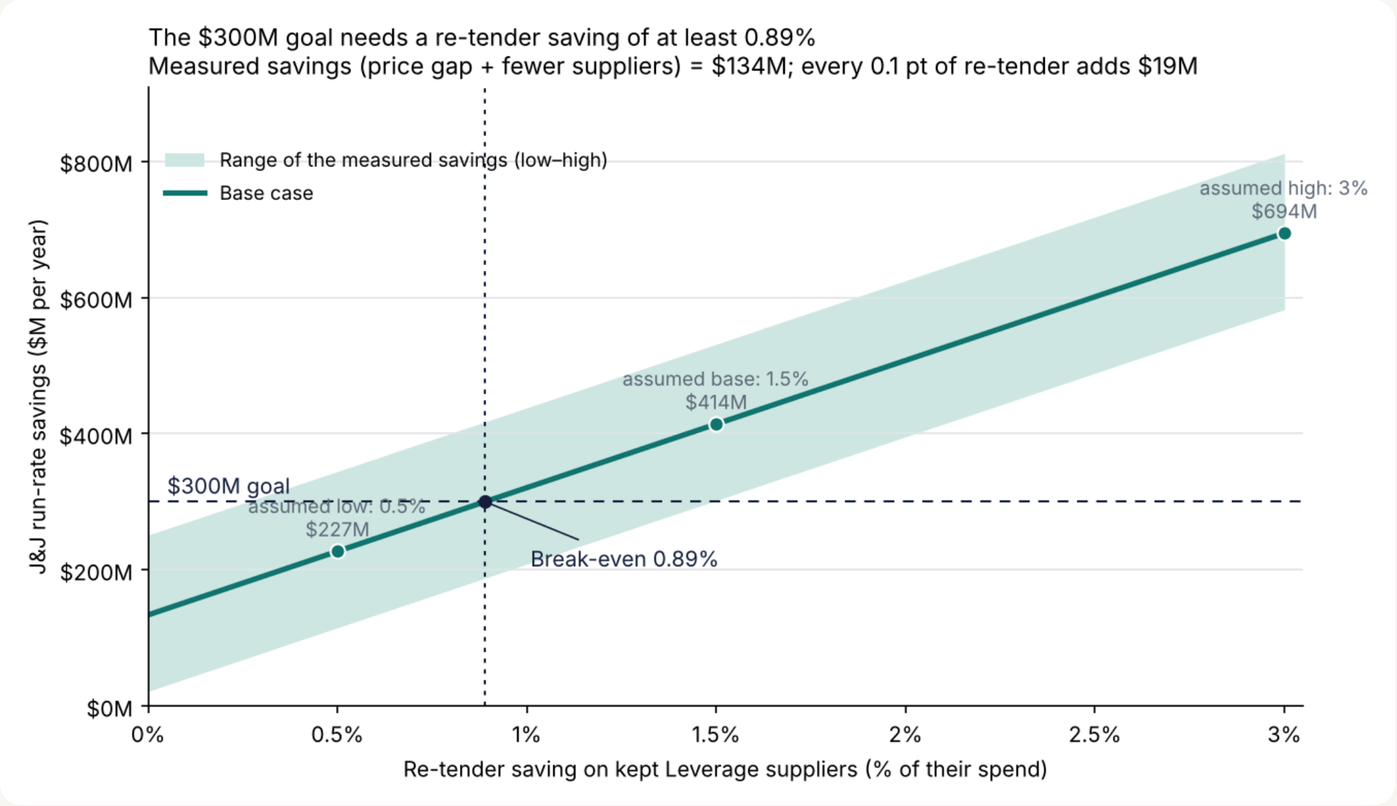
Exiting suppliers charged 9.8% more in Leverage and 5.2% more in Non-critical items, but were **cheaper** in Bottleneck (−10%) and Strategic (−7%). Savings only exist where the market is competitive.

To reach $300M, the re-tender of kept Leverage suppliers must save at least 0.89%.

Price-gap test: 127 goods items bought from both kept and exiting suppliers, median unit price, same UNSPSC and item

STEP 5B · HOW MUCH RESTS ON THE ASSUMPTION

The $300M goal needs a 0.89% re-tender saving



0.89%

break-even for $300M; 0.26–1.49% across the measured range

+$19M

a year for every 0.1 point of re-tender saving

Run-rate = $134M measured + re-tender rate × $18.7B kept Leverage spend at J&J scale · scripts/05b_retender_sensitivity.py

CHECKS ON THE CONCLUSION

Consolidation adds a little risk, and it is measured

Categories where one supplier holds > 50%

Quadrant	Before	After
All (406)	129	136
Strategic	9	9
Leverage	0	0
Bottleneck	95	99
Non-critical	25	28

10 newly cross 50% ($64M over 3 years): watch-list with a second-source clause.

Duplicate matching: ≈96% precise

70 pairs checked by hand: exact names 20/20, fuzzy 39/50. Fuzzy matches go to review before merging.

18 tests pass

No strategic or sole-source exits, a reason code on every row, totals reconcile raw → decisions.

Active spend, duplicates folded into their master · scripts/04c_concentration.py, 04d_duplicate_precision.py, tests/

PHASE 2 · RECOMMENDATION

From ~39,000 to ~32,100 with a category target count

The model, per category

1. Keep the fewest suppliers covering 90% of spend
2. At least 2 suppliers in Strategic and Bottleneck categories
3. Always keep strategic, sole-source and small/diverse suppliers
4. Move the rest to core suppliers or the catalog

Coverage	Cut	J&J
80%	−22%	~30,500
85%	−20%	~31,000
90%	**−18%**	**~32,100**
95%	−14%	~33,700

1.5% of spend changes supplier; dominated categories unchanged (161 of 382).

Run on the 4,580 California suppliers left after Phase 1 (FY2014–15), scaled to J&J’s ~39,000 · scripts/08_phase2_category_targets.py

REAL J&J SUPPLIERS · SEC FILINGS

The strategic tier is real, and concentration has a cost

Supplier	Supplies to J&J	Quadrant	Lesson for the plan
Alkermes	Exclusive maker of RISPERDAL CONSTA	Strategic	Sole source: never cut (R1), add MOU
MeiraGTx	RPGR gene therapy, 4+1-year supply deal (2023)	Strategic	Fixed term with extension option
Mersana	Clinical ADC manufacturing (2024)	Bottleneck	Tech-transfer right = second source
Emergent (former)	COVID-19 vaccine drug substance	Strategic	Ended 2022 over GMP breaches; $50M settlement

Of 9 candidates found by full-text search, 4 are confirmed suppliers. Legend is a partner, ORIC and Coherus buy from J&J, 2 are unconfirmed.

Sources: Alkermes, MeiraGTx, Mersana 10-K filings (SEC EDGAR); BioProcess International, Jul 2024 · docs/07_real_jnj_suppliers.md

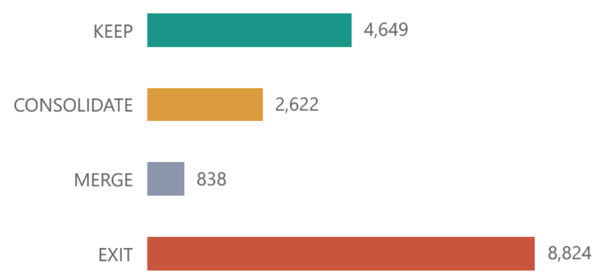
Dashboard · Executive summary

Supplier Rationalization: one decision and a reason code per supplier; 87% of spend stays with kept suppliers

Real procurement data: California purchase orders FY2012-15 (16,933 supplier records, 8,226 active in FY2014-15, $7.3B/yr), scaled to J&J: 80,000 suppliers reported earlier, ~39,000 today after the 2023 Kenvue spin-off

Suppliers today	Suppliers kept	Supplier reduction	Spend retained with kept suppliers	Small/diverse active spend retained	J&J run-rate savings, base ($M/yr)
16,933	4,649	72.5%	87.0%	96.6%	$414.1

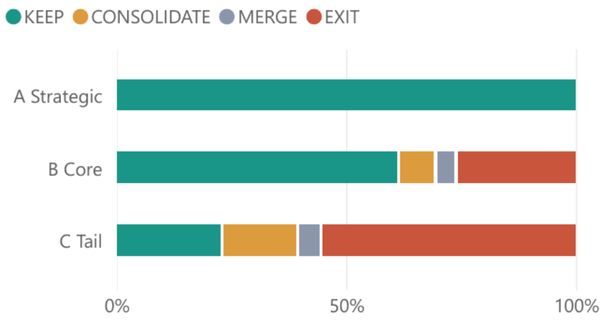
Suppliers by decision



Pareto: top 2% of suppliers hold 70% of spend



Decision mix by spend segment (A/B/C)



Every supplier has one decision and one reason code (rules applied in order R1-R8)

Rule	Decision	Reason	Sup
R1	KEEP	Strategic (A-segment) supplier	
R2	KEEP	Supply protection: top-3 active supplier in a Strategic/Bottleneck category, or the only active supplier in any category	
R3	MERGE	Duplicate record: fold into master supplier	
R4	EXIT	Inactive: no PO in 18+ months	
R5	EXIT	One-time vendor: 1 PO in 3 years; route future buys to preferred/catalog	
R6	KEEP	Preferred supplier: core of category spend or top-10% scorecard	
Total			

Spend segment

All

Kraljic quadrant

All

Small or diverse

All

Decision

All

So what

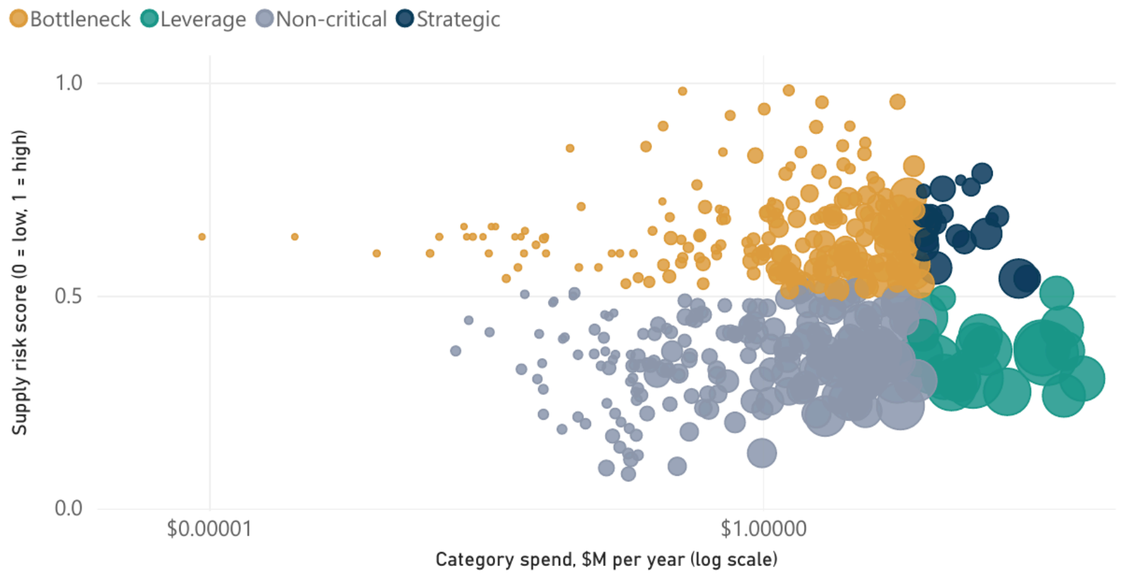
Small cut, big impact: a 27% cut (duplicates + one-time vendors) keeps 86% of the value.

Dashboard · Category strategy

Category strategy: consolidate leverage and non-critical categories, protect bottlenecks

Kraljic matrix of 409 real UNSPSC categories. Value = category spend; supply risk = non-competitive buying + top-3 concentration + thin supply base (ranked within size bands)

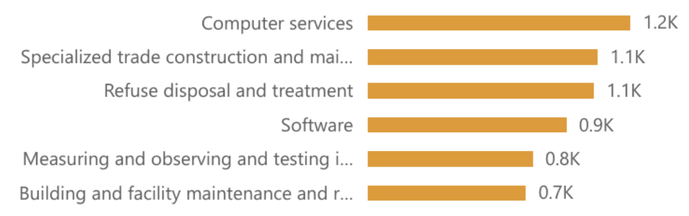
Kraljic matrix: bubble size = number of suppliers



Quadrant summary: 59% of spend sits in 20 leverage categories

Quadrant	Categories	Share of spend	Suppliers	Approach
Strategic	21	20.9%	3,231	Keep and partner: MC
Leverage	20	59.3%	12,663	Tender and consolida
Bottleneck	183	12.8%	10,341	Secure supply: long-t
Non-critical	185	7.0%	21,991	Simplify: catalog / pu
Total	**409**	**100.0%**	**48,226**	

Most fragmented categories: first consolidation targets



All 409 categories with quadrant and agreement type

Category	Quadrant	Agreement type	Spend $M/yr	Suppliers	% non-competitive	% top-
Professional engineering services	Leverage	Competitive tender; MSA with 2-3 preferred suppliers	$763.8	580	1.0%	
Comprehensive health services	Leverage	Competitive tender; MSA with 2-3 preferred suppliers	$539.3	400	1.4%	
Medical practice	Leverage	Competitive tender; MSA with 2-3 preferred suppliers	$521.5	518	0.5%	
Management advisory services	Leverage	Competitive tender; MSA with 2-3 preferred suppliers	$502.0	523	7.4%	
Human resources services	Leverage	Competitive tender; MSA with 2-3 preferred suppliers	$449.6	299	1.8%	

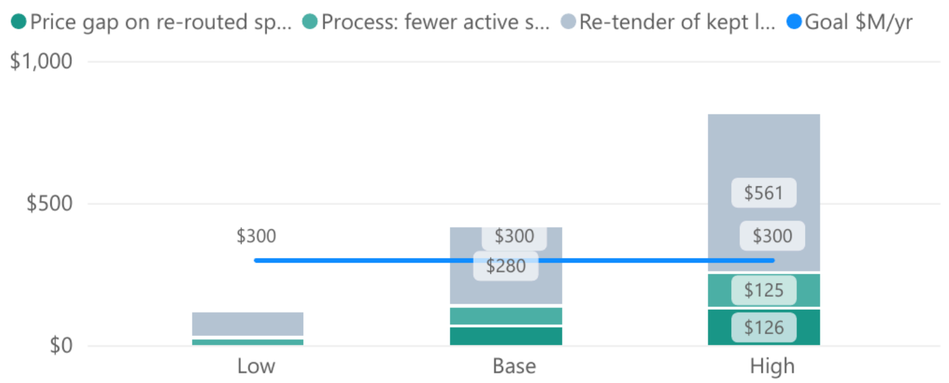
Dashboard · Savings case

Savings: $414M a year at J&J scale in the base case; $134M is measured, and a 27% cut already captures $356M

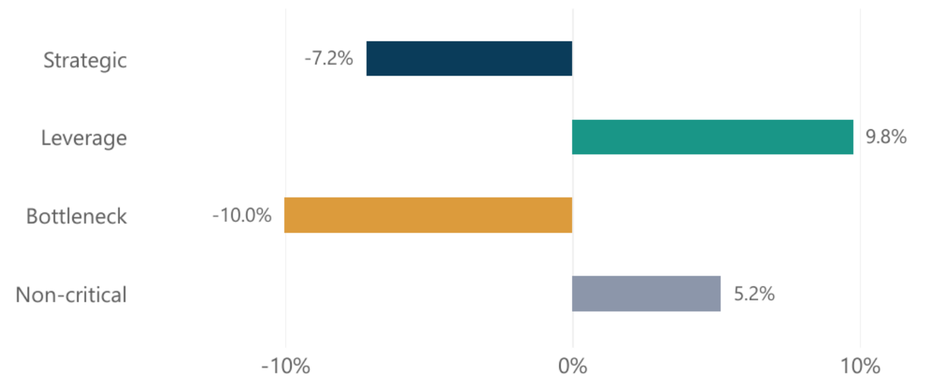
Price-gap test on 127 goods items bought from both kept and exiting suppliers; re-tender saving is an assumption. Scaled to J&J by spend ($35B) and supplier count (80,000 / 8,226 active suppliers in one year).

Run-rate savings, base ($M/yr)	Measured in the data ($M/yr)	One-time cost ($M)	3-year NPV, base ($M)	Re-tender saving needed for $300M goal
$414.1	$133.8	$73.4	$618.8	0.89%

J&J run-rate savings by scenario vs the $300M goal (line)



Exiting suppliers charge more only in Leverage (+9.8%) and Non-critical (+5.2%)



Price-gap evidence: same item, kept vs exiting suppliers

Item	Quadrant	Kept: median unit price	Exiting: median unit price	Gap	Exiting spend, 3 yrs
ammunition	Bottleneck	$265.45	$660.00	59.8%	$1,265,535
parts purchase for calfire aircraft	Bottleneck	$2,424.00	$1,804.00	-34.4%	$1,175,897
diesel fuel	Strategic	$5,026.59	$8,076.62	37.8%	$747,050
software maintenance	Leverage	$31,816.67	$39,703.00	52.4%	$712,936
propane	Bottleneck	$16,200.00	$10,000.00	-62.0%	$501,448
breast pumps	Bottleneck	$308,498.47	$162,736.00	-89.6%	$470,774

What has to be true

Measured savings: $63M price gap on re-routed spend + $71M fewer suppliers to manage.

The other $280M assumes a 1.5% re-tender saving with kept leverage suppliers; 152 of them hold 80% of that spend.

To reach $300M, the re-tender must deliver at least 0.89% (1.2% in the conservative plan): pilot it first.

One-time cost $73M base ($34M conservative); even the low case has a positive NPV ($158M).

Dashboard · Supplier explorer

Supplier explorer: decision and reason code for all 16,933 suppliers

Filter by decision, rule, segment or quadrant. Spend is annual (3-year total / 3).

Decision	Rule	Spend segment	Kraljic quadrant	Small or diverse
All	All	All	All	All

Supplier-level decisions

Supplier	Segment	Primary category	Quadrant	Decision	Rule	Reason
MANAGEMENT SOLUTION, LLC	A Strategic	Human resources services	Leverage	KEEP	R1	Strategic (A-segment) supp
Hensel Phelps Construction Co	A Strategic	Professional engineering services	Leverage	KEEP	R1	Strategic (A-segment) supp
CGI	A Strategic	Taxation	Strategic	KEEP	R1	Strategic (A-segment) supp
T.Y. Lin International	A Strategic	Professional engineering services	Leverage	KEEP	R1	Strategic (A-segment) supp
HP Enterprise Services	A Strategic	Comprehensive health services	Leverage	KEEP	R1	Strategic (A-segment) supp
Parsons Brinckerhoff, Inc.	A Strategic	Management advisory services	Leverage	KEEP	R1	Strategic (A-segment) supp
ACCENTURE	A Strategic	Software	Leverage	KEEP	R1	Strategic (A-segment) supp
A I M ENTERPRISES	A Strategic	Computer services	Leverage	KEEP	R1	Strategic (A-segment) supp
Intuitive Health Services	A Strategic	Medical practice	Leverage	KEEP	R1	Strategic (A-segment) supp
Hatch Mott MacDonald	A Strategic	Environmental management	Strategic	KEEP	R1	Strategic (A-segment) supp
HP Enterprise Services LLC	A Strategic	Management advisory services	Leverage	KEEP	R1	Strategic (A-segment) supp
Acs State & Local Solutions Inc.	A Strategic	Software	Leverage	KEEP	R1	Strategic (A-segment) supp
DynCorp International, LLC	A Strategic	Public order and safety	Strategic	KEEP	R1	Strategic (A-segment) supp
Sundt Construction	A Strategic	Professional engineering services	Leverage	KEEP	R1	Strategic (A-segment) supp
HealthRIGHT 360	A Strategic	Medical practice	Leverage	KEEP	R1	Strategic (A-segment) supp
Dataskill, Inc.	A Strategic	Software	Leverage	KEEP	R1	Strategic (A-segment) supp
Elavon, Inc. (formerly known as Nova Information Systems, Inc	A Strategic	Credit agencies	Strategic	KEEP	R1	Strategic (A-segment) supp
Pinnacle Petroleum	A Strategic	Fuels	Strategic	KEEP	R1	Strategic (A-segment) supp
Pinnacle Health Services, Inc	A Strategic	Human resources services	Leverage	KEEP	R1	Strategic (A-segment) supp
HDR Engineering, Inc.	A Strategic	Environmental management	Strategic	KEEP	R1	Strategic (A-segment) supp

STEP 7 · AGREEMENTS

One agreement type per quadrant for the suppliers we keep

Strategic · 417 suppliers

Partnership MOU: joint roadmap, quality agreement, executive reviews

Leverage · 1,649 suppliers

MSA term sheet: volume tiers and a benchmark clause to capture the ≥0.89% re-tender saving

Bottleneck · 1,061 suppliers

Long-term supply agreement with a second-source plan and tech-transfer right

Non-critical · 1,522 suppliers

Catalog and purchasing-card terms: standard prices, no negotiation cost

Plus transition notices for the 12,284 suppliers not kept (closure, consolidation, duplicate merge) · docs/agreements/

RECOMMENDATION AND ROADMAP

Cut the tail first, re-tender Leverage, protect the critical few

YEAR 1

Clean up

Merge duplicates, close inactive and one-time suppliers, launch catalogs

YEAR 1–2

Consolidate

Move R8 volume to preferred suppliers; re-tender Leverage under MSAs

YEAR 2–3

Secure

Strategic MOUs, second sources for Bottleneck items

KPI	Baseline	Year 3 target
Active suppliers	~39,000 today	~32,100 (Phase 2)
Run-rate savings (ramp 30 / 70 / 100%)	$0	$414M a year
Supply disruptions caused by exits	n/a	0

Gate before each phase: price-gap baseline met, no open quality issues, small/diverse spend ≥ 96%

APPENDIX

Sources and files

Data sources

- State of California Purchase Order Data 2012–2015 (data.ca.gov)
- J&J ESG reports and procurement disclosures (supplier count, spend)
- SEC EDGAR 10-K filings: Alkermes, MeiraGTx, Mersana, Legend, ORIC, Coherus, XBiotech
- BioProcess International, J&J–Emergent settlement, Jul 2024

Project files

- github.com/hemant-singla/jnj-supplier-rationalization
- Outputs workbook, one tab per step, Files_and_Sources tab
- Power BI dashboard, 4 pages, 21 DAX measures
- Scripts 01–07, SQL, agreement templates, business case